

# SME Strategic Valuation Model

A 4-Capital Framework for Italian Manufacturing

Design & Architecture · Methodology · Worked Example

## Project Scope

- Universe: 14,999 Italian SMEs · NACE 28xx (Machinery & Equipment manufacturing)
- Source: AIDA / Bureau van Dijk · 5 export files consolidated to 153-column dataset
- Output: Per-company valuation =  $\text{EBITDA} \times \text{Sector Multiple} \times \text{SQF} \times \text{GF}$
- Audience: Internal advisory · investment committee · academic review

# Executive Summary

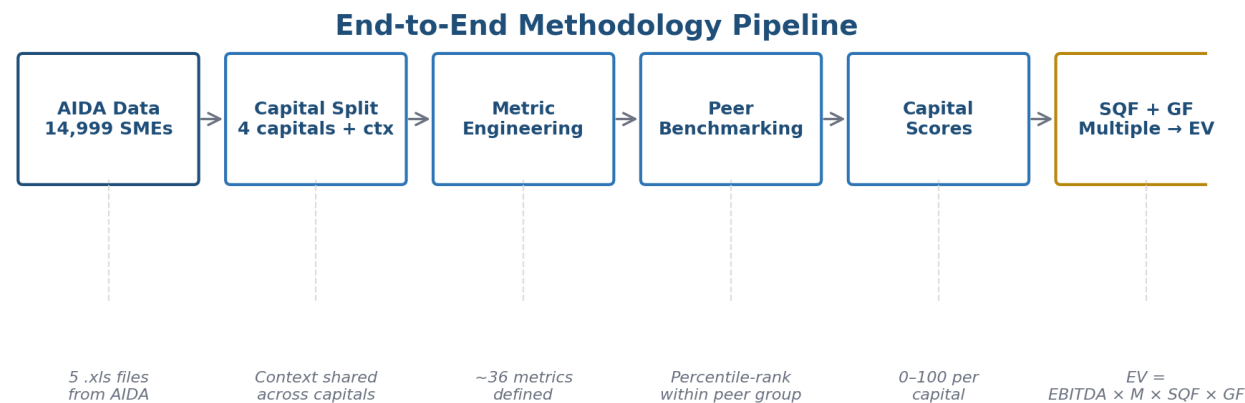
This document specifies a strategic valuation framework for Italian SMEs in the manufacturing sector (NACE 28xx). It translates 14,999 companies of AIDA financial and structural data into a transparent, peer-relative enterprise-value estimate.

Core valuation formula:

$$EV = EBITDA_{norm} \times M_{peer} \times SQF \times GF$$

where  $EBITDA_{norm}$  is a 3-year average,  $M_{peer}$  is the peer-group sector multiple, SQF is the Strategic Quality Factor (0.6–1.4), and GF is the Growth Factor (0.7–1.5). Companies in the same peer group share  $M_{peer}$ ; SQF and GF differentiate individual companies.

## Framework at a glance



## Five design principles

|               |                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------|
| Peer-relative | Every metric is normalised to a percentile within the company's AIDA peer group, not absolute thresholds.              |
| Multi-capital | Four lenses: Financial, Technological, Human & Organisational, Relational — weighted 35/20/25/20 for manufacturing.    |
| Transparent   | Linear aggregation, named fragility flags, defensible at each step. No black-box scoring.                              |
| Conservative  | Public-company multiples are explicitly discounted for illiquidity and small-cap before being applied to private SMEs. |
| Reproducible  | Built on the existing 6-file capital split. Adding a company only requires AIDA data; no manual scoring.               |

The remainder of this document specifies the metric catalogue, peer benchmarking approach, SQF and GF construction, sector multiple derivation, the medallion data architecture, and a worked example using **EUROCOIL S.R.L.**, a real cooling-equipment manufacturer drawn from the dataset.

# 1. Context & Scope

Italian SMEs are notoriously hard to value. Public-comparable transactions are sparse, ownership is closely held, and reported financials understate intangible value. AIDA provides the most comprehensive structured snapshot: financial statements, ATECO/NACE classification, peer-group assignment, and qualitative attributes (brands, alliances, internationalization).

This framework is restricted to **NACE 28 — Manufacture of machinery and equipment**. Single-sector scope simplifies the model in three ways: a single base sector multiple, homogeneous capital structures, and tighter peer-group benchmarks. The methodology can later be re-tuned for other sectors by re-weighting the four capitals and re-mapping the Damodaran industry source.

## Universe at a glance

|                       |                                                          |
|-----------------------|----------------------------------------------------------|
| Sector                | NACE 28 — Machinery & Equipment manufacturing            |
| Geography             | Italy (all provinces)                                    |
| Companies in dataset  | 14,999                                                   |
| Reporting period      | FY2022 – FY2024 (Last available year ≈ 2025 partial)     |
| Source                | AIDA / Bureau van Dijk · 5 separate exports consolidated |
| Final dataset shape   | 14,999 rows × 153 columns                                |
| Identifier columns    | 17 (company, peer group, ATECO/NACE codes)               |
| Capital-specific cols | 136 (40 fin · 40 tech · 33 H&O · 23 rel)                 |
| Peer groups           | AIDA-assigned, sub-sector + size band (e.g. 282 VL)      |

## Why a 4-capital model

Single-metric valuation (e.g. simple EBITDA multiple) ignores everything that distinguishes a fragile manufacturer from a resilient one. The 4-capital framework — financial, technological, human, relational — borrows from intellectual capital theory (Edvinsson, Stewart) but is grounded in observable AIDA fields rather than survey data. Each capital captures a distinct dimension of strategic quality:

|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Financial Capital      | How the business performs financially today: profitability, cash generation, leverage, liquidity. The dominant lens for any valuation.    |
| Technological Capital  | The hard infrastructure: plant, machinery, R&D capitalisation, IP, asset modernity. Most relevant for asset-heavy manufacturing.          |
| Human & Organisational | Workforce productivity, governance depth, management continuity, group structure, BvD independence. Critical for SME succession risk.     |
| Relational Capital     | Customer/supplier negotiating power, brand presence, internationalization, network membership, working capital cycle. Drives scalability. |



Figure 1. Capital weights are tuned for manufacturing — Technological is intentionally over-weighted vs. a services framework.

## 2. Methodology Overview

The framework processes data through six sequential stages. Each stage produces a named, inspectable artefact — never a single end-to-end black box.

|    |                           |                                                                                                    |
|----|---------------------------|----------------------------------------------------------------------------------------------------|
| 1. | <b>Ingestion</b>          | Read 5 AIDA exports, consolidate on tax code → 14,999 × 153.                                       |
| 2. | <b>Capital split</b>      | Split into 5 logical files: context, financial, technological, human & organisational, relational. |
| 3. | <b>Metric engineering</b> | Derive ~36 metrics from raw fields — ratios, CAGRs, stability indicators.                          |
| 4. | <b>Peer benchmarking</b>  | Convert each metric to a 0–100 percentile rank within the company's AIDA peer group.               |
| 5. | <b>Aggregation</b>        | Within-capital weighted sum → 4 capital scores (0–100). Across-capital weighted sum → CQS.         |
| 6. | <b>Valuation</b>          | Map CQS → SQF, compute GF, look up $M_{peer}$ , multiply through to indicative EV.                 |

### Why percentile ranking

Manufacturing financials are wildly heterogeneous: a small specialty parts maker and a very-large engineered-systems manufacturer can both legitimately sit in NACE 28 with very different absolute numbers. Three properties make percentile ranking the right primitive:

|                         |                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Outlier-robust</b>   | A single company with 60% EBITDA margin doesn't compress everyone else's score.                                                                             |
| <b>Sub-sector aware</b> | Comparing within the AIDA peer group means a 12% margin in a low-margin sub-sector scores high; the same margin in a high-margin sub-sector scores average. |
| <b>Interpretable</b>    | "Top quartile on profitability, bottom quartile on liquidity" is a sentence anyone can act on.                                                              |

## 3. Scoring Framework — Metrics per Capital

Each capital is built from 6 to 10 individual metrics. Within a capital, metrics are weighted to reflect their relative importance, with weights summing to 100%. All weights below are starting values — they should be tuned with sensitivity analysis in calibration.

### 3.1 Financial Capital **weight 35%**

Profitability, growth, leverage, liquidity, cash quality. The dominant capital — for any going-concern SME, financial performance is the floor on valuation.

| Metric                          | Weight | Source field(s)                  | Direction     |
|---------------------------------|--------|----------------------------------|---------------|
| EBITDA margin (3-yr avg)        | 15%    | EBITDA/Vendite % LY+24+23        | Higher better |
| EBITDA margin stability         | 8%     | 1 – CV(margin) over 3 yrs        | Higher better |
| Profit margin (3-yr avg)        | 8%     | Profit / Revenue                 | Higher better |
| Revenue CAGR 22→24              | 10%    | Revenues 24 vs 22                | Higher better |
| Current ratio                   | 10%    | Current ratio LY                 | Higher better |
| Debt / EBITDA                   | 12%    | Debt/EBITDA LY                   | Lower better  |
| Net financial position / Equity | 10%    | NFP / Total shareholder funds    | Lower better  |
| Cash conversion (CFO / EBITDA)  | 10%    | Cash flow ops / EBITDA           | Higher better |
| Equity ratio                    | 8%     | Shareholder funds / Total assets | Higher better |
| Asset turnover                  | 9%     | Revenue / Total assets           | Higher better |

### 3.2 Technological Capital **weight 20%**

Asset modernity, R&D commitment, IP intensity, investment trends. Manufacturing depends on tangible plant — but high intangibles signal differentiation, design rights, or proprietary know-how.

| Metric                                     | Weight | Source field(s)                                      | Direction                    |
|--------------------------------------------|--------|------------------------------------------------------|------------------------------|
| Capital intensity (Fixed assets / Revenue) | 10%    | TOTAL FIXED ASSETS / Revenue                         | Context-dependent†           |
| Plant & machinery growth 22→24             | 12%    | Plant & machinery 24 vs 22                           | Higher better                |
| R&D intensity                              | 15%    | R&D exp. / Revenue                                   | Higher better                |
| R&D growth 22→24                           | 10%    | R&D 24 vs 22                                         | Higher better                |
| IP intensity                               | 12%    | (Patents + Concessions/licenses) / Total intangibles | Higher better                |
| Tangible asset investment trend            | 12%    | Tangible investment LY (3-yr cum.)                   | Higher better                |
| Intangible asset investment trend          | 10%    | Intangible investment LY (3-yr cum.)                 | Higher better                |
| Asset balance (Tangible / Total fixed)     | 9%     | Tangible / Total fixed assets                        | Closer to peer median better |

|                                                                                                                                                                                |     |                               |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------|---------------|
| Industrial equipment intensity                                                                                                                                                 | 10% | Equipment / Plant & machinery | Higher better |
| <i>† Capital intensity is direction-context: extremely high values flag asset-bloat, extremely low flag asset-light competitors.<br/>Score is U-shaped around peer median.</i> |     |                               |               |

### 3.3 Human & Organisational Capital **weight 25%**

Workforce productivity, cost efficiency, governance depth, management continuity, ownership independence. Italian family-owned SMEs carry significant succession risk — this capital captures the people side that most other valuation frameworks miss.

| Metric                                  | Weight | Source field(s)                  | Direction            |
|-----------------------------------------|--------|----------------------------------|----------------------|
| Added value per employee                | 15%    | AV/employee LY                   | Higher better        |
| AV/employee growth 22→24                | 10%    | AV/employee 24 vs 22             | Higher better        |
| Turnover per employee                   | 10%    | Turnover/employee LY             | Higher better        |
| Turnover / Staff costs                  | 10%    | Turnover/Staff Costs LY          | Higher better        |
| Employee growth 22→24                   | 8%     | Employees 24 vs 22               | Modest positive best |
| Governance depth (directors / 100 emp.) | 10%    | N° directors / employees × 100   | Higher better        |
| Management continuity                   | 10%    | Current dir. / Previous dir.     | Higher better        |
| BvD independence indicator              | 9%     | A/B/C/D mapped to 100/75/50/25   | Higher better        |
| Group integration (companies in group)  | 10%    | log(N° companies in corp. group) | Higher better        |
| M&A activity flag                       | 8%     | Has M&A deals (binary)           | Higher better        |

### 3.4 Relational Capital **weight 20%**

Customer-supplier negotiating power, network embeddedness, brand presence, international reach. Where the company sits in its commercial ecosystem.

| Metric                               | Weight | Source field(s)                | Direction     |
|--------------------------------------|--------|--------------------------------|---------------|
| Customer power (avg credit days)     | 12%    | Durata media crediti LY        | Lower better  |
| Supplier leverage (avg payment days) | 10%    | Durata media debiti LY         | Higher better |
| Working capital cycle                | 10%    | Payment days – Credit days     | Higher better |
| Subsidiary network value             | 12%    | Subsidiary companies LY        | Higher better |
| Associated companies value           | 10%    | Associated companies LY        | Higher better |
| Internationalization (import/export) | 12%    | Import/export flag             | Yes better    |
| Foreign markets presence             | 10%    | Main foreign countries (count) | Higher better |
| Brand presence                       | 10%    | Main brand names (count)       | Higher better |
| Strategic alliances                  | 7%     | Strategic alliances flag       | Yes better    |
| Network membership                   | 7%     | Membership of network flag     | Yes better    |

## 4. Peer Benchmarking Approach

Peer groups are taken directly from AIDA's *Peer Group Name* field, which combines sub-sector (3- to 4-digit NACE) with a size band (Very Large, Large, Medium, Small). EUROCOIL S.R.L., for example, sits in **282 VL** — Very Large companies in NACE 282 (Manufacture of other general-purpose machinery), 1,174 members in AIDA, of which 482 are in our extracted dataset.

### Why use AIDA's peer groups instead of building our own

|                               |                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methodologically clean</b> | AIDA's peer groups are built from NACE classification + revenue band — the same logic an analyst would apply manually.                      |
| <b>Reproducible</b>           | AIDA peer assignment is deterministic and stable across exports. Custom k-means or quantile-bucket peer groups drift across data refreshes. |
| <b>Externally defensible</b>  | If a valuation is challenged, "AIDA peer group 282 VL" is a citable benchmark. "Cluster 7 from our PCA" is not.                             |

### Percentile-rank computation

$$\text{score}_i(c) = 100 \times \text{rank}_{\text{peer}}(c, m_i) / (n_{\text{peer}} - 1)$$

where  $m_i$  is metric  $i$ ,  $c$  is the company under valuation, and  $n_{\text{peer}}$  is the number of peers with valid data on that metric. For metrics where lower is better (Debt/EBITDA, credit days), the rank is inverted:  $\text{score} = 100 - \text{raw rank}$ .

### Capital aggregation: from metrics to capital scores

$$\text{CapitalScore}_k = \sum_{i \in k} w_i \times \text{score}_i(c) \quad \sum w_i = 1$$

### Composite Quality Score (CQS)

$$\text{CQS} = 0.35 \cdot \text{Fin} + 0.20 \cdot \text{Tech} + 0.25 \cdot \text{HumOrg} + 0.20 \cdot \text{Rel}$$

CQS is a single 0–100 number per company. It is the input to SQF, but is also useful in its own right as a comparative benchmark ("this firm scores 68 vs. peer average 50").

## 5. Strategic Quality Factor (SQF)

SQF translates the Composite Quality Score into a multiplicative valuation adjustment in the range **[0.6, 1.4]**. A neutral business — exactly at the peer median across all four capitals — has SQF = 1.0. Below 1.0 is a quality discount; above is a premium.

### Smooth mapping

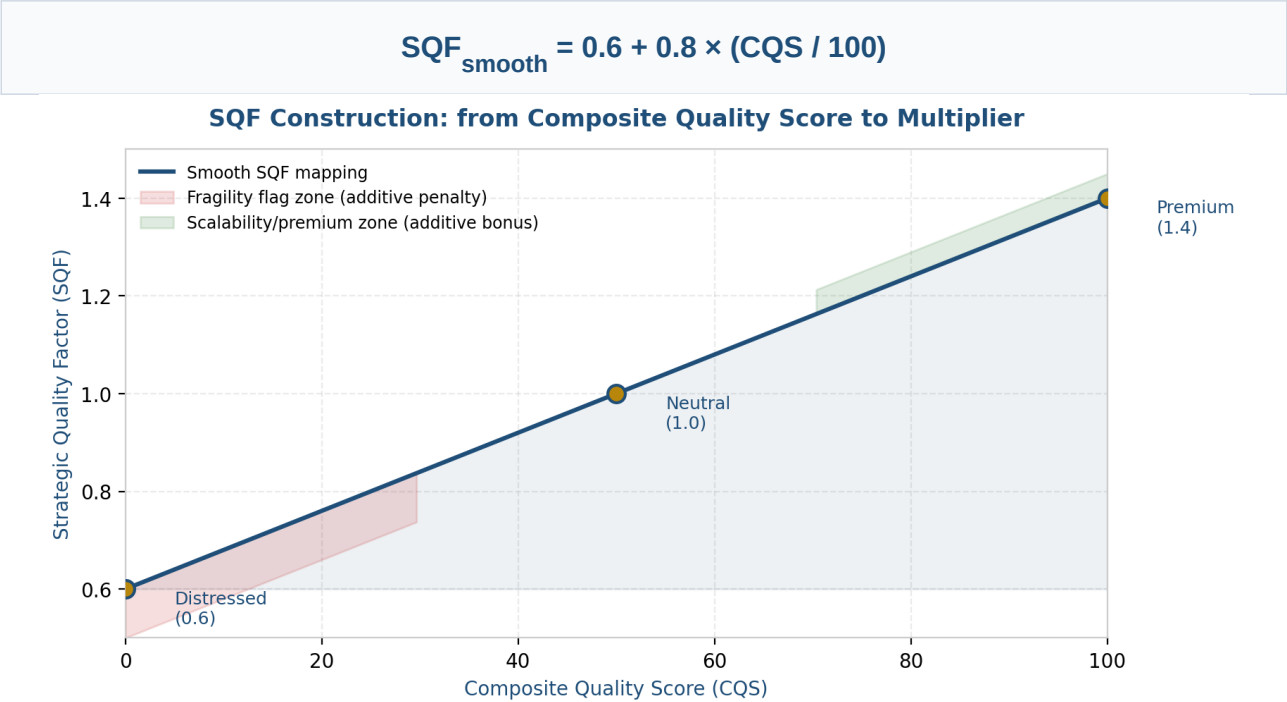


Figure 2. Linear smooth mapping with two non-linear adjustment zones.

### Fragility and scalability adjustments

The smooth mapping is overlaid with explicit additive adjustments that capture qualitative considerations the percentile machinery cannot see directly:

| Fragility flag           | Trigger                                        | SQF adj. |
|--------------------------|------------------------------------------------|----------|
| Excessive leverage       | Debt/EBITDA > 6 OR negative equity             | −0.10    |
| Liquidity stress         | Current ratio < 0.5                            | −0.05    |
| Persistent loss          | Profit < 0 in all 3 years                      | −0.10    |
| Negative cash generation | CFO < 0 in all 3 years                         | −0.05    |
| Governance thinness      | Single director, BvD = D                       | −0.05    |
| Scalability bonus        | Trigger                                        | SQF adj. |
| International scale      | Foreign markets ≥ 3 AND import/export = yes    | +0.05    |
| Brand & IP signal        | ≥ 2 brand names AND IP intensity > peer median | +0.05    |
| Productivity leadership  | AV/employee > peer 75th pct                    | +0.05    |

$$\text{SQF} = \text{clip}(\text{SQF}_{\text{smooth}} + \sum \text{adjustments}, 0.6, 1.4)$$

All adjustments are additive and the result is hard-clipped to [0.6, 1.4]. This keeps the framework bounded while letting qualitative signals shift a borderline business into discount or premium territory.

## 6. Growth Factor (GF)

SQF captures "how good is this business as it stands today". GF asks the orthogonal question: **what's its trajectory, and is that trajectory durable?** A firm with average current quality but consistent 15% organic growth deserves a premium on top of its SQF; a firm with the same headline growth driven by a single one-off acquisition does not.

### Three components

| Component                              | Definition                                                                                                                                                                                                               | Range              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| $g_{\text{norm}}$ — Peer-relative CAGR | Percentile rank of Revenue CAGR 22→24 within the peer group, divided by 100.                                                                                                                                             | [0, 1]             |
| $q$ — Quality-of-growth multiplier     | 1.00 if all three yearly growth rates (22→23, 23→24, 24→LY) are positive — i.e. <i>organic and consistent</i> . 0.85 if growth is concentrated in a single year ( <i>episodic</i> ). 0.70 if any year shows contraction. | {0.70, 0.85, 1.00} |
| $s$ — Size-class adjustment            | Slight upward tilt for smaller firms (Small/Medium peer-group size band) since they are expected to grow faster. Cancelled out for Very Large.                                                                           | [1.00, 1.05]       |

### Final formula

$$GF = \text{clip}(0.7 + 0.8 \times g_{\text{norm}} \times q \times s, 0.7, 1.5)$$

### Why distinguish organic from episodic growth

Italian SME revenue series often show step-function increases — typically from a single large contract win, an acquisition, or a covid-era catch-up year. A pure CAGR calculation treats this identically to consistent year-on-year compounding, but a buyer pays fundamentally less for episodic growth than for organic. The  $q$  multiplier formalises this: the same 14% CAGR scores  $GF = 1.42$  if it's three consecutive positive years, 1.27 if it's all from one year's spike, and 1.10 if any year shows contraction.

#### SQF and GF are deliberately orthogonal.

SQF answers: *how robust is this business?* → uses all four capitals.

GF answers: *how fast and how cleanly is it expanding?* → uses revenue series only.

Multiplying both lets a stable-but-stagnant firm and a fragile-but-rocketing firm reach similar valuations from opposite directions, which is correct.

# 7. Sector Multiple ( $M_{peer}$ )

All companies in the same AIDA peer group share the same  $M_{peer}$ . This is the right design: peer groups already encode sub-sector and size-band, which are precisely the attributes that drive systematic risk and therefore baseline multiples. Individual companies deviate on EV via SQF and GF — not by the multiple itself.

## Four-step build-up

| Step                                          | Source                                                                                                                                                      | Indicative impact |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| 1. Public-market base                         | Damodaran NYU industry EV/EBITDA dataset (free, quarterly), mapped from 2-digit NACE → Damodaran industry. For NACE 28 → "Machinery".                       | Anchor            |
| 2. Public→private discount                    | Illiquidity discount applied to public multiples for private SMEs. Empirical range 20–30% in the Italian middle-market literature (Mediobanca, KPMG).       | –25% (typical)    |
| 3. Italian SME size discount                  | Small-cap discount on top of illiquidity. Materially larger for Small/Medium peer-group bands; near-zero for Very Large.                                    | –10% (typical)    |
| 4. Sub-sector + peer-group quality adjustment | Bumps within ±10% based on (a) sub-sector deviation from the broad NACE 2-digit mean, and (b) peer-group aggregate profitability vs. broader manufacturing. | ±10%              |

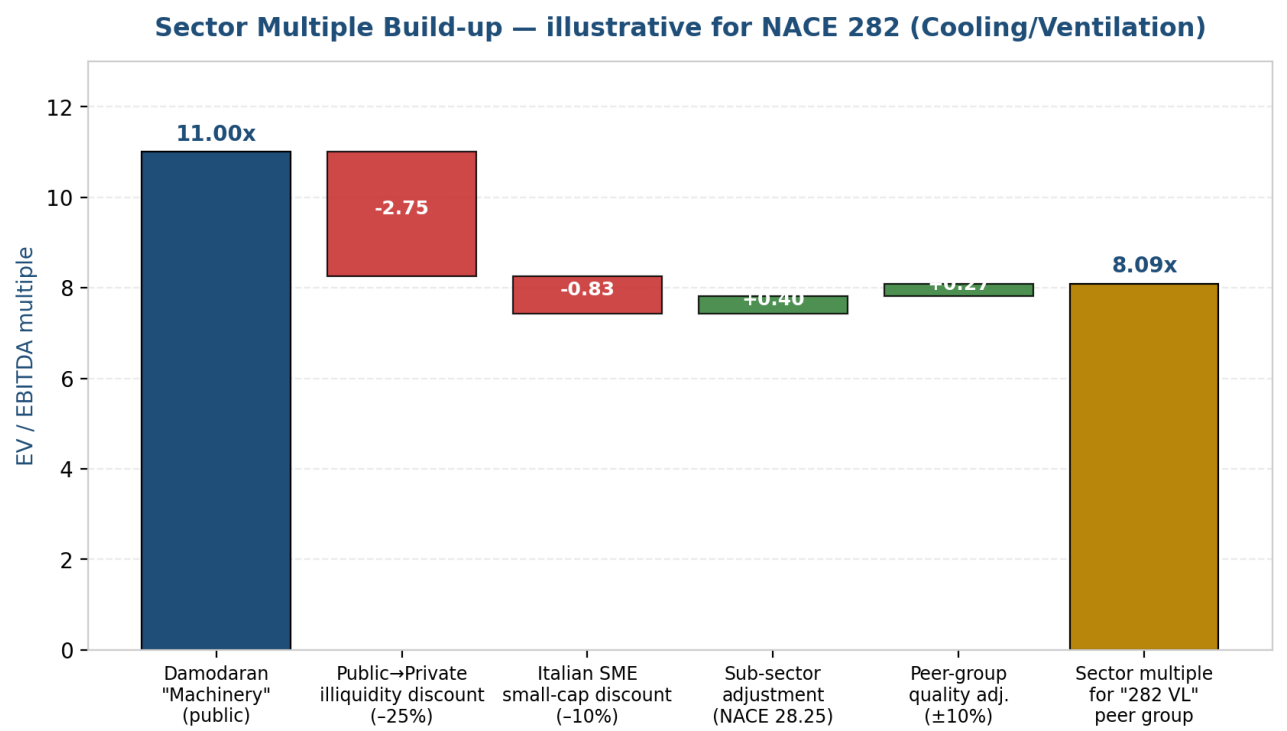


Figure 3. Illustrative build-up for the 282 VL peer group. The 8.09x landing point is a worked example, not a final calibration.

## Calibration sources

| Source                            | Use                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| Damodaran NYU Stern industry data | Public-company EV/EBITDA by industry — the base anchor. Updated quarterly, freely available. |

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|                                                |                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Mediobanca "Le società italiane" annual report | Italian large-mid manufacturing financial benchmarks. Calibration check on private discount magnitude.                           |
| KPMG Italian M&A reports                       | Observed transaction multiples for closed deals in Italian manufacturing. The most relevant external check.                      |
| AIDA peer-group internal financials            | Used only for the relative peer-group quality adjustment (step 4), never as the absolute multiple anchor — to avoid circularity. |

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## 8. Final EV Formula

EV = EBITDA<sub>norm</sub> × M<sub>peer</sub> × SQF × GF

| Term                   | Definition                                                                                                                        | Range / unit |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------|
| EBITDA <sub>norm</sub> | 3-year average EBITDA (LY, 2024, 2023). Smooths one-off years and reduces the leverage of bumper or trough years on the multiple. | € / ke / Me  |
| M <sub>peer</sub>      | Peer-group sector multiple as defined in §7. Same value for every company in the same AIDA peer group.                            | Multiple (×) |
| SQF                    | Strategic Quality Factor as defined in §5. Captures how the company is running across all 4 capitals.                             | [0.6, 1.4]   |
| GF                     | Growth Factor as defined in §6. Captures both speed and quality of growth.                                                        | [0.7, 1.5]   |

### How wide is the resulting range

Combined SQF × GF spans **[0.42, 2.10]**. That means at the same EBITDA × M<sub>peer</sub>, the framework can value a top-quintile organic grower at roughly **5×** the EV of a bottom-quintile fragile contractor. This spread is consistent with observed transaction multiples in Italian manufacturing M&A: "good" companies trade at 8–10× EBITDA, "distressed but viable" at 2–4×.

**Equity value bridge.** EV is the operating-asset value. To go from EV to Equity Value for an actual transaction, subtract Net Financial Position and add cash and non-operating assets. For benchmarking and ranking, EV is sufficient and avoids capital-structure noise.

## 9. Worked Example — EUROCOIL S.R.L.

EUROCOIL S.R.L. is a real Italian manufacturer drawn from the dataset — a Verona-based producer of finned coils for air and primary fluid heat exchange (NACE 28.25 — non-domestic cooling and ventilation equipment). The company has 199 employees, fiscal year ending 30 September 2025, and is a member of a 92-company corporate group.

### Profile snapshot

|                         |                                                                            |
|-------------------------|----------------------------------------------------------------------------|
| Tax code                | 03819650379                                                                |
| Province                | Verona                                                                     |
| NACE 4-digit            | 28.25 — Non-domestic cooling and ventilation equipment                     |
| Peer group              | 282 VL — Very Large companies in NACE 282 (1,174 members; 482 in our data) |
| Employees               | 199 (was 129 in 2022 → +54% in 3 years)                                    |
| Revenue trajectory (€k) | 37,295 (2022) → 43,867 (2023) → 45,477 (2024) → 54,783 (LY)                |
| EBITDA trajectory (€k)  | 6,959 (2022) → 9,240 (2023) → 7,549 (2024) → 10,148 (LY)                   |
| EBITDA margin           | 18.6% → 20.9% → 16.4% → 18.3% (top-quartile but volatile)                  |
| Net financial position  | €18.2M (positive = net debt)                                               |
| Debt / EBITDA           | 1.83 (improving from 4.19 in 2022)                                         |
| Current ratio           | 0.80 (low — working-capital-intensive)                                     |
| BvD independence        | D (member of corporate group, ultimate owner identifiable)                 |

### Peer percentile rank — actual values from the dataset

Computed against the 482 members of the 282 VL peer group present in our data. Higher = better; metrics inverted where appropriate.

| Metric                   | EUROCOIL value            | Peer percentile | Verdict                         |
|--------------------------|---------------------------|-----------------|---------------------------------|
| EBITDA margin (LY)       | 18.3%                     | 80th            | Top-quintile profitability      |
| EBITDA margin (2024)     | 16.4%                     | 75th            | Strong                          |
| Debt / EBITDA*           | 1.83×                     | 26th            | Higher leverage than most peers |
| Current ratio            | 0.80                      | 1st             | Severe liquidity gap            |
| AV per employee          | €102,850                  | 62nd            | Above peer median               |
| Turnover per employee    | €278,240                  | 41st            | Slightly below median           |
| Staff cost per employee* | €51,850                   | 75th            | Cost-efficient labor structure  |
| Turnover / Staff costs   | 5.37                      | 55th            | Average                         |
| Revenue CAGR 22→24       | +10.4% (peer median 4.9%) | 96th            | Above-average growth            |

\* lower-is-better metric (rank inverted)

EUROCOIL S.R.L. — Peer Percentile Radar (282 VL group, n=482)

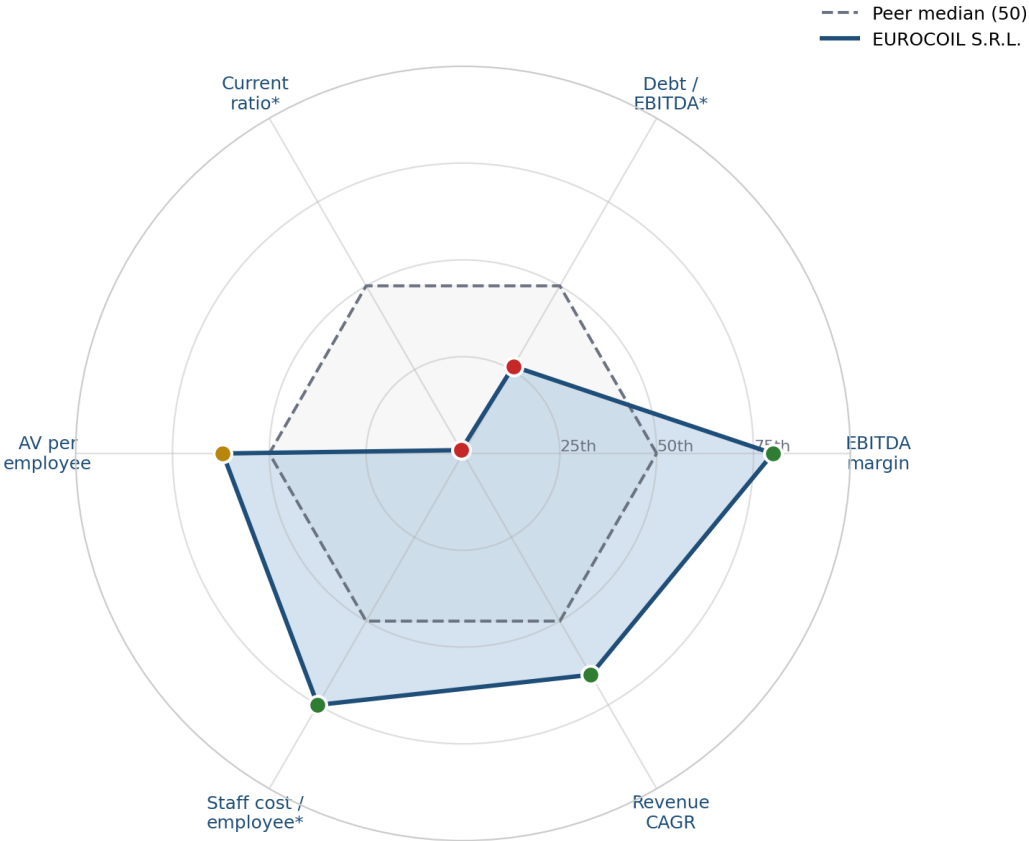


Figure 4. EUROCOIL's actual peer percentile profile within group 282 VL. The asymmetry — strong on the right (margins, cost, growth), weak on the top-left (liquidity, leverage) — is exactly the kind of nuance the framework is designed to surface.

## 9.1 Indicative scoring & valuation

Using the percentile data above plus illustrative weights and adjustments. These are indicative numbers to demonstrate the mechanics — full calibration requires running the metric pipeline across all 36 metrics, not just the 9 sampled here.

| Step                                 | Computation                                                                                                             | Result      |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------|
| Financial Capital score              | Profitability strong (75–80) but liquidity (1) and leverage (26) weak → weighted average                                | ≈ 52 / 100  |
| Technological Capital score          | Plant & machinery growth strong (+60% 22→24); R&D nil; intangible holdings notable                                      | ≈ 55 / 100  |
| Human & Org. Capital score           | Productivity above median; group-embedded (92-company group); independence indicator D                                  | ≈ 62 / 100  |
| Relational Capital score             | Limited international presence (Import/export = no, no foreign markets); but improving credit days; large group context | ≈ 38 / 100  |
| <b>Composite Quality Score (CQS)</b> | $0.35 \times 52 + 0.20 \times 55 + 0.25 \times 62 + 0.20 \times 38 = \mathbf{52.4}$                                     | <b>52</b>   |
| SQF (smooth)                         | $0.6 + 0.8 \times 52/100 = 1.02$                                                                                        | 1.02        |
| Fragility adjustment                 | Liquidity flag (current ratio < 0.5 → no, but stretched); no other flags trigger                                        | −0.05       |
| Scalability bonus                    | Productivity ≥ 75th pct (AV/emp 62nd → does not trigger). No bonuses fire.                                              | 0.00        |
| <b>SQF (final)</b>                   | $\text{clip}(1.02 - 0.05, 0.6, 1.4)$                                                                                    | <b>0.97</b> |
| Revenue CAGR percentile              | 66th → $g_{\text{norm}} = 0.66$                                                                                         | 0.66        |
| Quality of growth                    | All 3 yearly growths positive (+18%, +4%, +20%) → q = 1.00 (organic, consistent)                                        | 1.00        |
| Size adjustment                      | Very Large peer group → s = 1.00                                                                                        | 1.00        |
| <b>GF</b>                            | $0.7 + 0.8 \times 0.66 \times 1.00 \times 1.00 = 1.23$                                                                  | <b>1.23</b> |
| Sector multiple (282 VL)             | From \$7 build-up — illustrative                                                                                        | 8.09×       |
| Normalized EBITDA (3-yr avg)         | $(10,148 + 7,549 + 9,240) / 3$                                                                                          | €8.98M      |

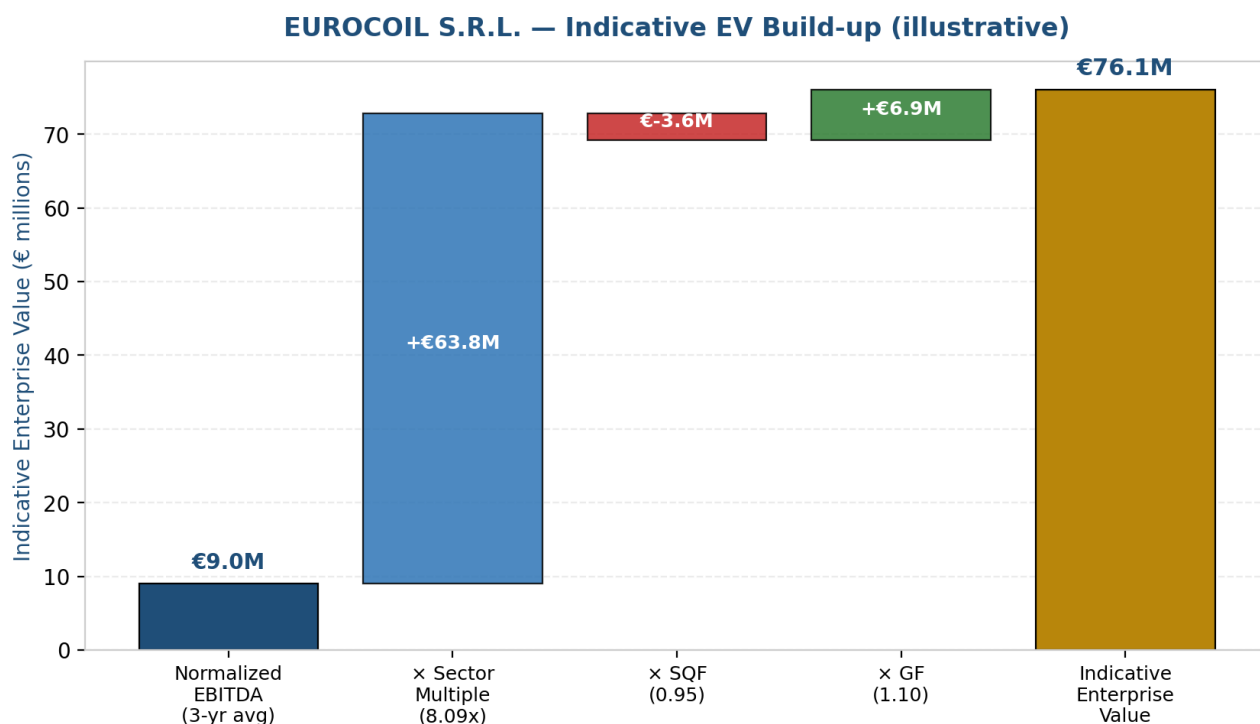


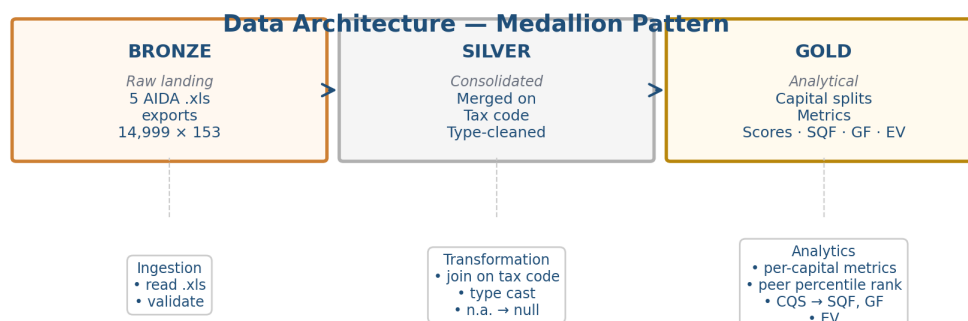
Figure 5. EV bridge for EUROCOIL — from normalized EBITDA to indicative Enterprise Value.

**Indicative EV = €8.98M × 8.09 × 0.97 × 1.23 ≈ €87M**

Reading: a peer-group neutral business at the same EBITDA would value at €72.6M; EUROCOIL's strong organic growth (GF = 1.23) more than offsets its modest financial-quality discount (SQF = 0.97), yielding a ~20% premium over the peer-neutral baseline.

## 10. Data Architecture

The pipeline follows a medallion pattern (Bronze / Silver / Gold) — a layered approach that separates raw landing, cleaned consolidation, and analytics-ready outputs. Each layer is idempotent and individually queryable; failures at any stage do not corrupt earlier layers.



Outputs: context\_columns.xlsx · financial\_capital.xlsx · technological\_capital.xlsx · human\_organisational\_capital.xlsx · relational\_capital.xlsx · all\_capitals\_clean\_split.xlsx · scores.parquet · valuations.csv

Figure 6. Medallion data architecture. Each layer is independently reproducible from the layer above.

### Layer responsibilities

|                              |                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bronze — Raw Landing</b>  | Five .xls files from AIDA stored verbatim. No transformation. The ground truth.                                                                                  |
| <b>Silver — Consolidated</b> | Single 14,999 × 153 Parquet/xlsx. Joined on tax code, types coerced (n.a. → null, tax codes preserved as strings with leading zeros), year ordering normalized.  |
| <b>Gold — Analytical</b>     | Per-capital tables, computed metrics, peer-rank tables, capital scores, SQF, GF, $M_{peer}$ , EV — all in one star-schema-style Parquet store keyed on tax code. |

### Existing artefacts

The Bronze and most of the Silver layer are already in place from the data engineering work preceding this design:

| File                              | Layer  | Purpose                                                       |
|-----------------------------------|--------|---------------------------------------------------------------|
| AIDA_Consolidated.xlsx            | Silver | Joined master 14,999 × 153                                    |
| context_columns.xlsx              | Silver | Identifier columns only                                       |
| financial_capital.xlsx            | Silver | Context + 40 financial columns                                |
| technological_capital.xlsx        | Silver | Context + 40 technological columns                            |
| human_organisational_capital.xlsx | Silver | Context + 35 human / governance columns                       |
| relational_capital.xlsx           | Silver | Context + 23 relational columns                               |
| all_capitals_clean_split.xlsx     | Silver | Combined workbook with 5 sheets                               |
| scores.parquet (planned)          | Gold   | Per-company metric values + percentile ranks + capital scores |
| valuations.parquet (planned)      | Gold   | Per-company SQF, GF, M, EV                                    |

# 11. Implementation Roadmap

| Phase                         | Deliverable                                                                                                                                    | Status     |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 0 · Data foundation           | Consolidated 14,999 × 153 dataset; capital splits; column catalog.                                                                             | Done       |
| 1 · Metric engineering        | Implement ~36 derived metrics (ratios, CAGRs, stability indicators) as Python functions. Output: <i>metrics.parquet</i> , one row per company. | Next       |
| 2 · Peer benchmarking engine  | Group-by Peer Group Name → percentile-rank each metric → invert lower-is-better. Output: <i>peer_ranks.parquet</i> .                           | Next       |
| 3 · Capital score aggregation | Apply within-capital weights → 4 capital scores. Apply across-capital weights → CQS.                                                           |            |
| 4 · SQF and GF computation    | Smooth mapping + fragility/scalability flags + clip. GF using revenue series.                                                                  |            |
| 5 · Sector multiple lookup    | Build $M_{peer}$ table: Damodaran base × discounts × peer-group adjustment. One value per AIDA peer group.                                     |            |
| 6 · EV computation & ranking  | Apply final formula. Produce per-company valuation table + peer-relative ranking.                                                              |            |
| 7 · Sensitivity analysis      | Tornado charts on capital weights, fragility-flag thresholds, multiple discount magnitude.                                                     | Validation |
| 8 · Reporting layer           | Per-company one-pager generator (peer radar + score breakdown + EV bridge). Streamlit or static-PDF export.                                    | Stretch    |

# 12. Limitations & Sensitivity

## Acknowledged limitations

| Limitation                                                         | Mitigation                                                                                                              |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| AIDA reporting lag (LY ≠ same calendar year for all companies)     | Use "Last available year" + 2024/2023/2022 explicit columns. Date-stamp each company.                                   |
| Missing data on qualitative fields (brands, alliances, customers)  | Treat missing as a separate signal — "not reported in AIDA" itself indicates relational-capital opacity.                |
| Sector multiple anchoring on Damodaran (US public companies)       | Cross-check against Mediobanca and KPMG Italian transaction data. Recalibrate quarterly.                                |
| Peer group size variance (some groups have 50 members, some 1,500) | Set minimum peer group size of 30 for benchmarking. Below threshold → fall back to NACE 3-digit + size band.            |
| CFO data sparsely populated ("n.a." common)                        | Cash conversion metric weight reduced when CFO unavailable; weight redistributed to other financial metrics.            |
| Single-sector scope (NACE 28 only)                                 | Capital weights and fragility thresholds will need re-calibration per sector. Current weights are manufacturing-tilted. |

## Sensitivity priorities

Three parameters drive the bulk of valuation variance and warrant explicit sensitivity in Phase 7:

- |                                     |                                                                                                                                                        |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Capital weights</b>           | 35/20/25/20 for Fin/Tech/H&O/Rel is the proposed starting point. Test $\pm 10$ pp shifts and observe rank-order stability across the 14,999 companies. |
| <b>2. Public-private discount</b>   | 25% baseline; literature ranges 20–35%. A 10pp swing here moves every EV by $\sim 10\%$ .                                                              |
| <b>3. Fragility flag thresholds</b> | Debt/EBITDA > 6, current ratio < 0.5, etc. These create discontinuities — sensitivity should verify no large clusters sit exactly on a threshold.      |

# Appendix A. Symbol Reference

| Symbol                 | Meaning                                                 |
|------------------------|---------------------------------------------------------|
| EBITDA <sub>norm</sub> | 3-year average EBITDA, smoothed                         |
| M <sub>peer</sub>      | Sector multiple specific to the peer group              |
| CQS                    | Composite Quality Score (0–100)                         |
| SQF                    | Strategic Quality Factor (0.6–1.4)                      |
| GF                     | Growth Factor (0.7–1.5)                                 |
| g <sub>norm</sub>      | Peer-relative CAGR percentile (0–1)                     |
| q                      | Quality-of-growth multiplier (organic vs. episodic)     |
| s                      | Size-class adjustment for GF (1.00–1.05)                |
| score <sub>i</sub> (c) | Percentile rank of metric i for company c (0–100)       |
| w <sub>i</sub>         | Weight of metric i within its capital ( $\Sigma = 1$ )  |
| EV                     | Enterprise Value (operating-asset value)                |
| LY                     | Last available year (varies by company fiscal calendar) |

# Appendix B. Capital Weights — Starting Calibration

| Capital                | Weight | Rationale (manufacturing context)                                                                                  |
|------------------------|--------|--------------------------------------------------------------------------------------------------------------------|
| Financial              | 35%    | Profitability and cash generation are the floor of any going-concern valuation. Heaviest weight is non-negotiable. |
| Human & Organisational | 25%    | Italian SMEs are people-intensive; succession risk and management depth materially move private valuations.        |
| Technological          | 20%    | Manufacturing is asset-heavy. R&D and IP intensity differentiate within-sub-sector winners.                        |
| Relational             | 20%    | Customer power, internationalization, brand presence drive scalability — but harder to measure from AIDA alone.    |

# Appendix C. Glossary of AIDA Field Conventions

| Field convention      | Meaning                                                                     |
|-----------------------|-----------------------------------------------------------------------------|
| <i>Last avail. yr</i> | Most recent fiscal year filed — varies by company (some end Sept, some Dec) |
| <i>th EUR</i>         | Thousands of Euro (i.e. multiply value by 1,000 to get Euros)               |

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|                                 |                                                                                                                                                                                   |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>n.a.</i>                     | Not available — company did not report or value not derivable. Treated as null in scoring.                                                                                        |
| <i>EBITDA/Vendite</i>           | Italian for EBITDA-to-Sales ratio; equivalent to EBITDA margin                                                                                                                    |
| <i>Durata media dei crediti</i> | Average days sales outstanding (DSO) — receivable collection period                                                                                                               |
| <i>Durata media dei debiti</i>  | Average days payable outstanding (DPO) — supplier payment period                                                                                                                  |
| <i>Peer Group</i>               | AIDA-assigned cluster combining 3-digit NACE + size band (XS / S / M / L / VL)                                                                                                    |
| <i>BvD independence</i>         | Bureau van Dijk indicator: A (independent), B ( $\geq 1$ known shareholder $< 25\%$ ), C ( $\geq 1$ known shareholder $\geq 25\%$ ), D ( $\geq 1$ known shareholder $\geq 50\%$ ) |

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